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Laser-induced wet stability and adhesion of pure conducting polymer hydrogels

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Conductive hydrogels can be used to make electrodes that interface with biological tissues due to their similar mechanical properties and high electrical conductivity in physiological environments. The electrical and mechanical properties of conductive hydrogels have improved in recent years, but they still suffer from poor durability and reliability, particularly in wet environments. Here we show that high-stability conductive hydrogels can be fabricated and adhered to various substrates using laser-induced phase separation and interface structures. With this approach, conducting polymers can be selectively transformed into conductive hydrogels with wet conductivities of 101.4 S cm^{-1} and patterned with a spatial resolution down to 5 µm. The conductive hydrogels exhibit high robustness, maintaining their electrochemical properties after 1 h of ultrasonication and 8 months of storage in water. They also exhibit peel and lap-shear strength in wet conditions of 64.4 N m^{-1} and 62.1 kPa, respectively. We used the conductive hydrogels to make microelectrode arrays that can stably record electrophysiological signals over 3 weeks in rat brains and hearts. The hydrogel electrodes can also be reused through intensive ultrasonication cleaning due to their durability.

Creating soft interfaces between electronic devices and biological systems is important for applications such as neural implants[1], health monitoring devices[2] and augmented reality applications[3]. Numerous soft electronic materials have been developed to achieve this. Conductive hydrogels[4] are of particular interest as their mechanical properties are closer to biological tissues than other materials (such as conductive composite elastomers[5] and metal percolation networks[6]). Most conductive hydrogels are fabricated by incorporating conductive fillers (such as metal nanomaterials[7], graphene[8] or conducting polymers[9]) in crosslinked non-conductive hydrogel backbones.

Conductive hydrogels that use the conducting polymer poly (3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS) as a filler have been intensively studied due to their high electrical conductivity, softness and electrochemical stability. However, electronics using such soft conducting polymer-based hydrogels are limited by their short-term operation due to their poor robustness in wet environments and their easy delamination from the substrates due to interfacial water penetration, which eventually causes the devices to fail. In addition, since the hydrogel backbones for mechanical support in physiological environments are electrically non-conductive, the conductivity is very low (below 10 S cm^{-1})[10–12]. As a result, conductive

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hydrogels with both mechanical robustness and high electrical conductivity have been developed using pure conducting polymers, without a non-conductive supporting backbone[13–16].

Conductive hydrogels made with pure conducting polymer show high mechanical stability and electrical conductivity in wet conditions, but the delamination issues from the substrates remain unresolved. To overcome this, increasing the roughness of the substrate surface[17,18] and introducing thin adhesive layers[19,20] have been explored to improve their wet stability and adhesion. However, increasing the roughness of the substrate requires complicated fabrication processes and is unsuitable for soft and stretchable substrates, whereas adding a non-conductive adhesive layer can lower the conductivity. Increasing the adhesion force by incorporating self-adhesive additives has also been shown to sacrifice electrical conductivity[21]. Therefore, an approach for making pure conducting polymer hydrogels with high robustness and strong adhesion to substrates is required for the stable long-term operation of bio-interfacing electronics in wet physiological environments. In addition, understanding the coordinated electrophysiological signals from biological systems such as the brain and heart requires that many micrometre-sized cells are recording simultaneously[22]. Therefore, high-resolution patterning techniques for soft conductive hydrogels also need to be developed to produce several recording sites with a high density of arrays for the limited size of biological systems.

In this article, we report a laser-induced process for stably adhering pure PEDOT:PSS hydrogel to polymer substrates. The resulting samples have high conductivity and softness in wet physiological environments. This process allows patterning of the hydrogel with micrometre-scale resolution. A selective photothermal reaction between the PEDOT:PSS and the polymer substrate is induced using a continuous-wave 532 nm laser. The laser is scanned through the transparent substrate and is absorbed at the interface of micrometre-thick PEDOT:PSS, which leads to partial phase separation in the PEDOT:PSS (ref. 15) and the formation of micro- and nanoscale interlockings on the interface. We then dip the PEDOT:PSS in ethylene glycol (EG), a typical solvent for phase separation, which strengthens the interconnections between the PEDOT-rich domains.

Our laser-induced phase separation and adhesion (LIPSA) method can directly bond PEDOT:PSS to various polymer substrates, including polyethylene terephthalate (PET), polyimide (PI), thermoplastic polyurethane (TPU), styrene-butadiene-styrene (SBS) and styrene-ethylene-butylene-styrene (SEBS). Finally, the structure is immersed in water, which transforms the bonded PEDOT:PSS into conductive pure PEDOT:PSS hydrogels containing high amounts of water (over 80% water content) for implantable and wearable electronics. The pure PEDOT:PSS hydrogel exhibits a high wet electrical conductivity of up to 101.4 S cm^{-1} due to the reconstruction of its polymer networks by laser-induced phase separation. It can maintain impedance and charge-storage capacity (CSC) even after 1 h of sonication (ultrasonic power 100 W), which illustrates its high mechanical robustness and electrical stability. The measured wet adhesion forces between the PEDOT:PSS hydrogel and the substrate show a peel strength of 64.4 N m^{-1} and lap-shear strength of 62.1 kPa.

The laser scanning can also be used to functionalize PEDOT:PSS at the substrate interface in selected areas. This allows the PEDOT:PSS hydrogel to be directly patterned with a spatial resolution down to 5 µm while maintaining its robust wet stability. We show that pure PEDOT:PSS hydrogel microelectrode arrays can be used for long-term in vivo neural recording in rat brains, recording electrophysiological signals stably for 3 weeks. Hydrogel microelectrodes can also stably record electrophysiological cardiac signals from dynamically beating rat hearts. After the recording periods, the hydrogel microelectrodes are robust enough to be cleaned using harsh ultrasonication, making them suitable for reusable, soft, bio-interfacing electronics. (A comparison of our process with previous studies is shown in Supplementary Table. 1 (refs. 13–16,23–26)).

Laser-induced wet adhesion of pure PEDOT:PSS on substrates

Many studies of rigid PEDOT:PSS have been reported with thicknesses down to the nanoscale to reduce the effective bending stiffness and interfacial pressure on tissues[27]. However, this severely deteriorates its electrical and electrochemical properties, resulting in low signal quality from biological systems. Also, these ultrathin electrodes are, typically, fragile and crumple easily. Due to the dominance of van der Waals forces, it is hard to restore the devices and they require intense surgical handling techniques for real-world clinical applications[14,28]. On the other hand, soft PEDOT:PSS hydrogels can resolve the mechanical mismatch with interfacing biological tissues by lowering the Young's modulus and by making water-rich environments[4]. Therefore, fabricating PEDOT:PSS hydrogels with microscale thickness is more effective for realizing a high signal quality from tissues and for easy surgical applicability in clinical applications.

To successfully develop high-resolution PEDOT:PSS hydrogels with enough thickness, we optimized our laser parameters by considering the optical properties of PEDOT:PSS and the substrates. Thin PEDOT:PSS has a high transparency, but as the thickness increases, the transparency decreases and it changes from blue to dark black, resulting in the high absorption of visible light (Supplementary Fig. 1). Polymer substrates currently used in implantable and wearable bioelectronics are developed to be transparent in the visible wavelength range[29]. By irradiating a 532 nm visible laser beam onto the front side of PEDOT:PSS, the surface regions experience the maximum temperature. Due to the low thermal conductivity of pristine PEDOT:PSS, heat dissipates inside micro-thick PEDOT:PSS and cannot be sufficiently transferred to the substrate, which fails to form a strong bond with the PEDOT:PSS. However, by scanning the laser over the surface of the substrate (backside scanning), a 532 nm laser can be transmitted through the transparent polymer substrates. The interfacing PEDOT:PSS then absorbs the light to generate highly concentrated photothermal energy at the interface, which produces strong bonds. We simulated the heat transfer to confirm this hypothesis and calculated the temperature distribution as the laser was scanned while changing the irradiation direction (see the detailed information in Supplementary Note 1). The temperature of the substrate interfacing with the PEDOT:PSS rose by more than twice for backside scanning compared to front-side scanning. This can create a thermodynamic environment suitable for changing the interfacial structure (Supplementary Fig. 2).

To apply this physical phenomenon, we flipped the samples so that the transparent polymer substrate, which transmits the 532 nm visible laser, was on top (Supplementary Fig. 3) and the thick PEDOT:PSS film (dried state) was on the bottom. We then irradiated the laser onto the interface (Fig. 1a(i)). Before laser irradiation, pristine PEDOT:PSS has a phase distribution such that the core is electrically conductive and hydrophobic PEDOT and the shell surrounding the PEDOT is insulating and hydrophilic PSS (Fig. 1a(i), dotted box). When the PEDOT:PSS and polymer substrate are scanned by the laser, as reported in our group's previous study[15], 'phase separation' occurs. The photothermal reaction substantially increases the interconnection between PEDOT-rich domains (Fig. 1a(ii)). Simultaneously, the thermal energy that is concentrated between the polymer substrate and the PEDOT:PSS creates mechanical interlocking structures and spot welding between the substrate and PEDOT:PSS, thus dramatically increasing the interfacial bonding strength (Fig. 1a(ii), dotted box). The PEDOT:PSS is then soaked in water to wash away the untreated area, which completely dissolves in the water due to the surrounding water-dissolvable PSS shell. The part scanned by the laser remains on the substrate due to phase separation and the increased bonding strength caused by mechanical interlocking and spot welding (Fig. 1a(iii)). However, the distribution of the laser absorption is not spread evenly in the depth direction inside the PEDOT:PSS, so the degree of phase separation is weaker further from the interface (Fig. 1a(iii), dotted box). This originates from the

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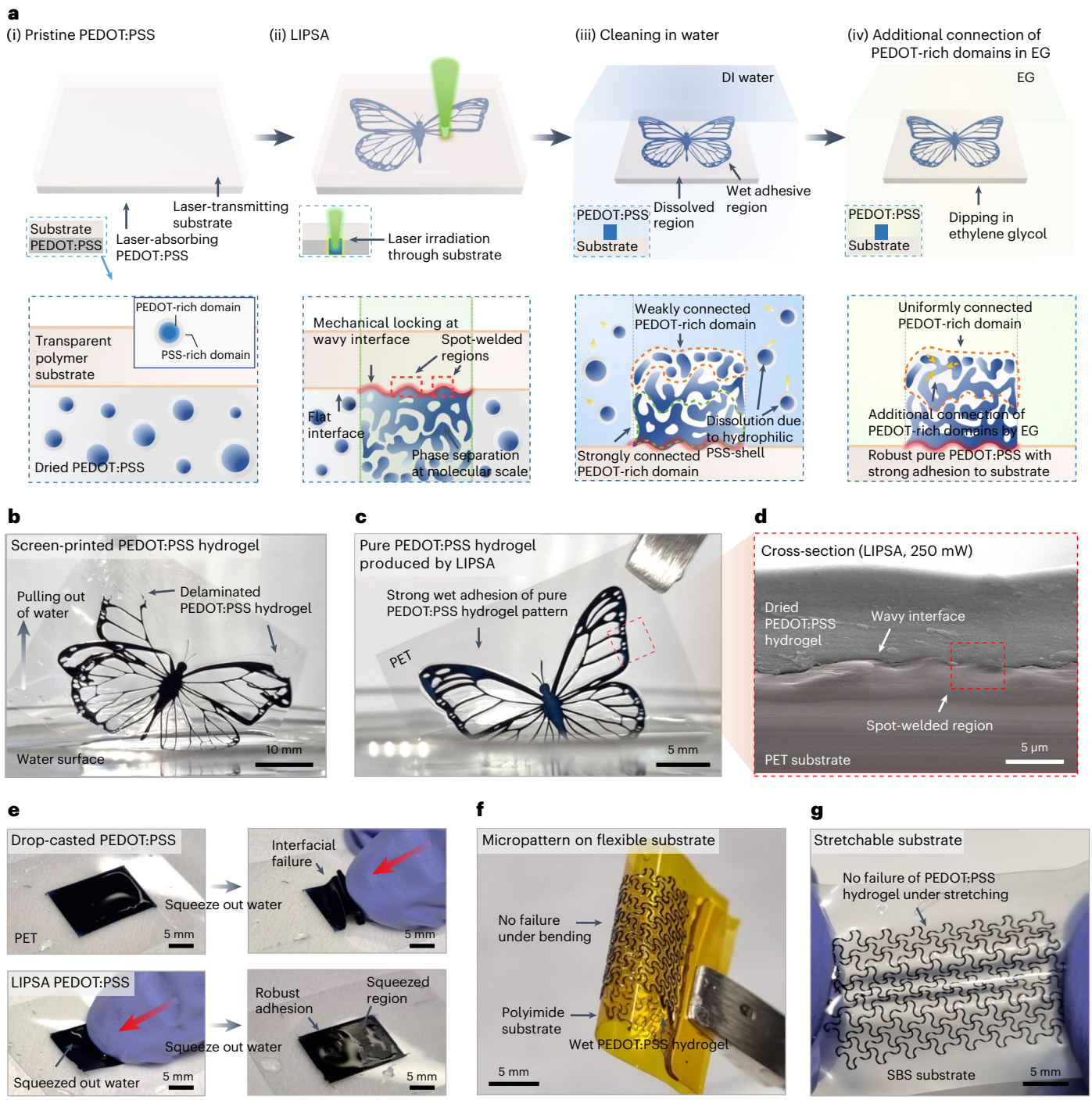


Fig. 1 | Laser-induced strong wet stability and adhesion of pure PEDOT:PSS hydrogels on diverse substrates. a, Schematic illustration of LIPSA. **b**, Digital image of screen-printed PEDOT:PSS hydrogels produced by EG treatment, showing its weak adhesion to the substrate. **c**, Pure PEDOT:PSS hydrogel produced by LIPSA, showing its robust adhesion to the substrate in wet conditions. **d**, SEM image of the cross-section of dried PEDOT:PSS hydrogel on a PET substrate, showing wavy interlocking interface structures and spot-welded zones. **e**, Comparison of wet adhesion to the substrate under a shear force. Top, samples treated only with EG. Bottom, LIPSA-treated samples. **f**,**g**, Micropatterned PEDOT:PSS hydrogels on a flexible PI substrate (**f**) and a stretchable SBS substrate (**g**). The micropatterned hydrogel adhered strongly to the substrate under mechanical deformation. DI water, deionized water.

relatively thick depth and inefficient heat transfer inside PEDOT:PSS. Thus, we indirectly estimated the relative degree of phase separation within PEDOT:PSS through the heat transfer simulation by calculating the spatial temperature distribution under laser irradiation (Supplementary Note 1). Due to the concentration of photothermal energy at the interface, the strongest phase separation occurs at the interface of the transparent polymer substrate and weakens in the thickness direction. Sufficient phase separation can reach as far as 72% of the total thickness in PEDOT:PSS, and irregular, partial, phase separation may occur elsewhere (Supplementary Fig. 4).

Moreover, to fabricate completely robust and adhesive PEDOT:PSS hydrogels, post-treatment with EG is used to further strengthen the interconnections of the weakly connected PEDOT-rich domains (Fig. 1a(iv)). Transmission electron microscopy (TEM) observations

of a laser-treated PEDOT:PSS surface before EG treatment revealed laser-induced expanded and irregular phase separation of PEDOT:PSS. Relatively large aggregates of PEDOT-rich domains formed on the top surface (Extended Data Fig. 1a). This enlarged PEDOT-rich domain enables fast charge transport and improved wet stability in physiological environments. However, as a trade-off for the large aggregation of PEDOT-rich domains, samples treated only with a laser show scattered aggregates of PSS. This irregularity appears due to the attenuated photothermal energy that heats PEDOT:PSS more unevenly in the depth direction. PSS-rich domains are weakly in contact with each other through hydrogen bonds, and since hydrogen bonds are easily destroyed by moisture, aggregates of PSS act as a defect that lowers mechanical properties such as adhesion strength and wet stability when moisture is absorbed. Additional connections of PEDOT-rich domains were made through the EG post-treatment. We verified with TEM images of EG-treated PEDOT:PSS surfaces that regions with PSS agglomeration were no longer present and that the aggregation of the PEDOT-rich domains became smaller and uniform (Extended Data Fig. 1b). Although laser and EG treatment enhanced the connectivity of PEDOT-rich domains, they did not enhance the crystallinity. The (100) and (200) direction peaks are clearly evident in a grazing-incidence wide-angle X-ray scattering profile[30], grazing-incidence X-ray diffraction spectra[24] and X-ray diffraction data[31] of sulfuric acid-treated PEDOT:PSS, indicating a highly ordered structure. However, distinct peaks were not observed in the X-ray diffraction data of laser-treated PEDOT:PSS before and after EG treatment (Extended Data Fig. 2).

Through our optimized laser process combined with EG post-treatments, the micropatterned pure PEDOT:PSS hydrogels made by LIPSA showed superior robustness in a wet environment compared to conventional patterning techniques (for example, screen printing and mould-casting) (Fig. 1b,c). Patterned hydrogels maintained their stability even after 8 months in wet environments due to strengthened PEDOT-rich connections and multiscale interfacial structures (mechanical interlocking and spot welding), as shown in scanning electron microscope (SEM) images (Fig. 1d and Supplementary Fig. 5). Pure PEDOT:PSS hydrogel fabricated by LIPSA is highly robust and stable in wet conditions; even when it is in a wet state and water is squeezed out, a stable shape is preserved with strong bonds to the substrate (Fig. 1e and Supplementary Video 1). On the other hand, although PEDOT:PSS hydrogel treated only with EG is robust, an interfacial failure was triggered almost immediately by the external squeezing force (Supplementary Video 2). Our laser process is not limited only to a specific substrate but can be widely applied to substrates that are universally used for wearable and implantable electronics, such as PET, PI, TPU, SBS and SEBS (Fig. 1f,g and Supplementary Fig. 6). As a demonstration, micropatterned PEDOT:PSS hydrogels on TPU substrates showed robust wet adhesion, which endured various types of mechanical stresses (repetitive rubbing with fingers, stretching, twisting, bending and crumpling) (Supplementary Videos 3 and 4).

Phase separation and its effects on interfacial water

The conductivity of PEDOT:PSS hydrogel in the dried state increased dramatically from 0.03 to 394 S cm^{-1} due to the laser-induced phase separation. Subsequently, the conductivity further increased to 543 S cm^{-1} after enhancing the connectivity of PEDOT through EG treatment (Fig. 2a and Supplementary Fig. 7). As the laser power was increased, the phase separation occurred more strongly, leading to an increase in the conductivity. Under the optimal condition of 250 mW, the conductivity of PEDOT:PSS hydrogel in a fully swollen state reached 101.4 S cm^{-1}. However, at 280 mW, the excessive laser energy carbonized the PEDOT:PSS, resulting in a decrease in conductivity (Fig. 2b). The conductivity in both the wet and dry states was determined by measuring the thickness of the PEDOT:PSS hydrogel using an optical microscope. The PEDOT:PSS hydrogel (250 mW) showed excellent hygroscopicity with 81.6% water content (Supplementary Fig. 8) and anisotropic swelling (Supplementary Fig. 9). The high pattern resolution is the most notable advantage of the laser process. LIPSA, using a Galvano mirror scanner system, was able to form a pattern with a width of 20 μm and a thickness of 25 μm (highest aspect ratio 1.25) in a fully swollen state (Fig. 2c). By using a high-magnification objective lens system with a tighter focus, it was possible to form a pattern with a width of 5 μm and a thickness of 2 μm (Extended Data Fig. 3, Supplementary Fig. 10 and Supplementary Note 2).

PEDOT:PSS hydrogel consists of two regions in the thickness direction. The water-absorbing region was less affected by the laser. The water-inhibiting region is in direct contact with the transparent polymer substrate, where strong phase separation occurred upon direct laser irradiation, resulting in the formation of a hydrophobic PEDOT-rich domain. Moisture generally weakens the interfacial bonding force between the two materials[32]. Indeed, the bonding mechanism between PET and PEDOT:PSS is known to be hydrogen bonding[33], which is vulnerable to moisture. Thus, the water-inhibiting region acts as a wet adhesion promoter, hindering water penetration at the interface between PEDOT:PSS and the polymer substrate (Fig. 2d). The difference in the degree of phase separation between the two regions is evident in the phase images captured by atomic force microscopy (AFM). Relatively small PEDOT grains are surrounded by PSS in the water-absorbing region, whereas strongly connected PEDOT is separated from the PSS to form large grains in the water-inhibiting region (Fig. 2e). The peak for PSS in the water-inhibiting region as seen by X-ray photoelectron spectroscopy was smaller than that of PSS in the water-absorbing region, indicating that the water-inhibiting region is a PEDOT-rich domain compared to the water-absorbing region (Supplementary Fig. 11).

We carried out all-atom molecular dynamics (MD) simulations to gain a theoretical understanding of the compositional dependency of PEDOT:PSS as an inhibitor layer of water penetration. From a thermodynamic perspective, our focus was on how well the environment, where phase separation occurs at the nanoscale due to laser irradiation, can inhibit water intrusion. The composition range of PSS to PEDOT was from 0:8 to 80:4 based on the number of molecules (0:1 to 0.48:0.52 wt%). The maximum fraction of PEDOT was determined by referring to ref. 34. After placing a water layer on the modelled polymer domains, the interaction energy density was calculated after the domains had penetrated the bulk region. The computational details are provided in Supplementary Note 3 (Supplementary Fig. 12 and Supplementary Table 2).

Figure 2f shows the difference in interaction force with absorbed water for each composition. This difference represents the level of blocking of moisture intrusion, which depends on the degree of phase separation in PEDOT:PSS. The absolute value of the interaction energy density formed by the unit water molecules is much higher with hydrophilic PSS than with hydrophobic PEDOT. One notable finding is that the magnitude of the interaction energy density between PSS and water is not constant and decreases with the advent of PEDOT. This suggests that the intrinsic hydrophobic properties of PEDOT oligomers within a PEDOT:PSS system directly interfere with the water permeation properties of the surrounding PSS. Specifically, as shown in Fig. 2g,h, we found that the sulfonated group of PSS is preoccupied with the adjacent PEDOT. Such a morphological feature of the PEDOT:PSS system retards the diffusion of water into the interior and ensures the high strength of the interface. If the permeated water is to form a hydrogen bond with PSS, the ionic bond between PEDOT and PSS must first be broken. The effect is more pronounced as the content of PEDOT in the system increases, making water molecules unfavourable in largely phase-separated PEDOT:PSS regions, which coincides with the experimental results.

Wet adhesion measurement

The robust wet adhesion of LIPSA was evaluated quantitatively by performing 90° peel-off and lap-shear tests. The wet adhesion of

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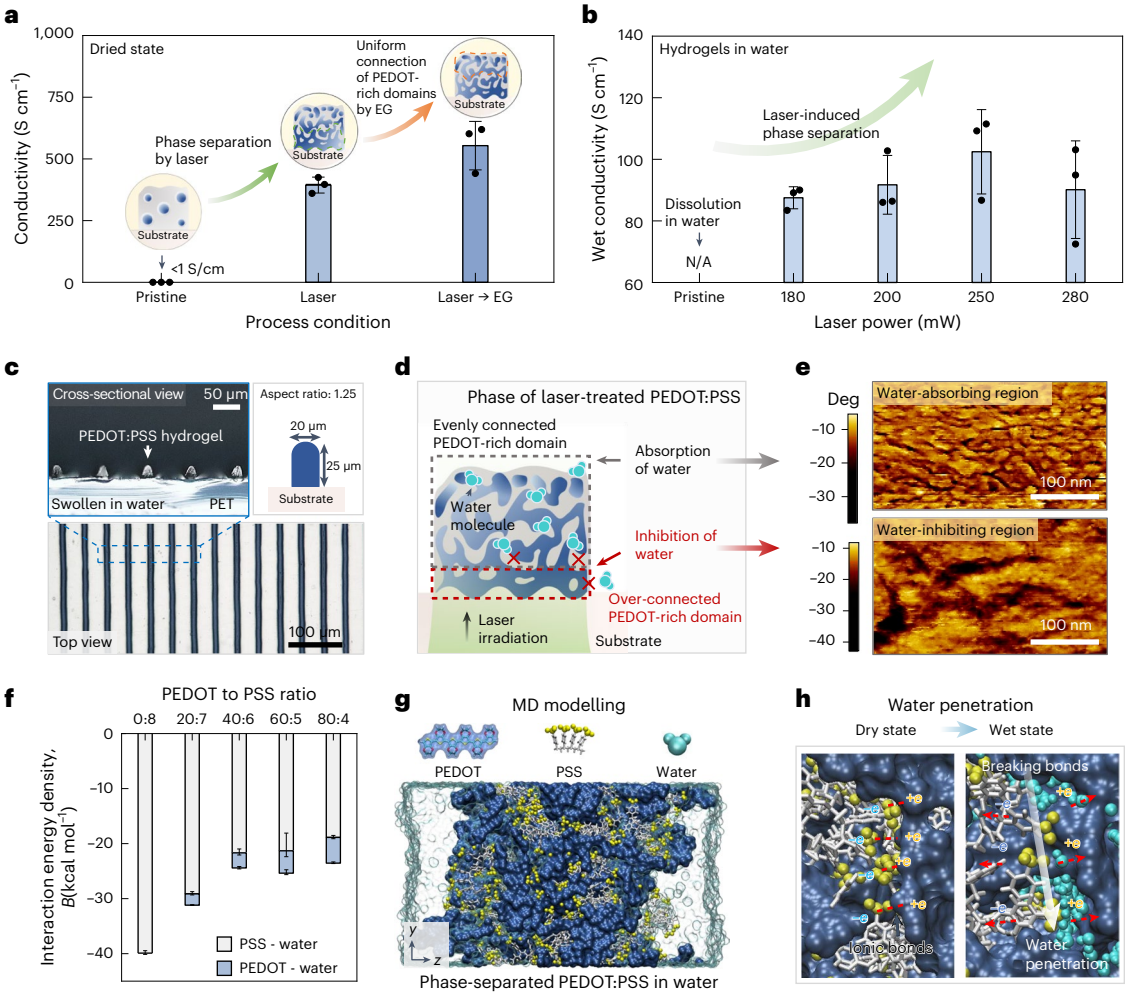


Fig. 2 | Phase separation and its effect on interfacial water penetration. **a**, Conductivity of laser-treated PEDOT:PSS hydrogel at each fabrication stage. **b**, Electrical conductivity of PEDOT:PSS hydrogel according to laser power. **c**, Pattern resolution of LIPSA (20 μm width and aspect ratio 1.25). Cross-sectional view (top) and top view (bottom). **d**, Schematic illustration of the difference in the degree of phase separation (Deg), which varies in the thickness direction. **e**, AFM phase images of water-absorbing region (top) and water-inhibiting region (bottom). **f**, Interaction energy density calculated through simulations based on the PEDOT to PSS ratio. **g**, Set-up for MD modelling. **h**, Breaking of ionic bonds between PEDOT and PSS due to water penetration. Values in **a** and **b** represent the means ± standard deviation ($n = 3$).

bulk-annealed PEDOT:PSS hydrogels was also measured as a control for the laser-treated samples. A peel-off test measures the resistance to fracture in which the opening mode (mode I) and sliding mode (mode II) are mixed, but for a high peel-off angle, the fracture mode can be assumed to be pure mode I (ref. 35). The lap-shear test measures the resistance to a pure mode II fracture. Figure 3a shows the set-up and specimen used in the 90° peel-off test to measure wet adhesion between PET and the PEDOT:PSS hydrogel. Due to the low modulus (less than 15 MPa) and elongation of the swollen pure PEDOT:PSS hydrogels (Supplementary Fig. 13), they were not strong enough to withstand the peel force and tore. Therefore, the test was conducted by peeling the PET substrate from the PEDOT:PSS hydrogel in water, which was attached to a glass substrate with epoxy adhesive. The representative peel-off strength of each sample was determined by dividing the average value of the measured load, excluding the first peak load, by the contact width (Fig. 3b). After the first load peak, for which an excessive force was applied to open the crack, the peel force stabilized and there was a plateau. The measured peel-off strengths were plotted as functions of laser power and bulk-annealing temperature (Fig. 3c). Figure 3d shows the set-up and specimen used in the lap-shear test. Wet adhesion between PET and PEDOT:PSS hydrogels was measured by pulling the PET substrate and a water-permeable nylon membrane filter attached to the PEDOT:PSS hydrogels. A pipette provided additional water to keep the specimen swollen. The nylon membrane filter was hydrophilic and compatible with PEDOT:PSS hydrogels. Its high stiffness allowed it to withstand high loads without deformation[19]. The lap-shear strength is the shear stress at the moment of failure, and the maximum shear stress was determined as the representative shear strength of each specimen (Fig. 3e). The measured lap-shear strengths were also plotted as functions of laser power and bulk-annealing temperature (Fig. 3f). The wet adhesion strengths against mode I and mode II fractures have the same trend (Fig. 3c,f). For the bulk-annealed specimens, wet adhesion increased at low adhesion levels with increasing annealing temperature, being 12.4 N m^{-1} and 14.3 kPa on average at 100 °C annealing temperature. As the laser power was increased in LIPSA, the wet adhesion dramatically increased, being 61.5 N m^{-1} and 56.0 kPa on average at 250 mW of laser power. The optimal wet adhesion was at 250 mW; when a higher power was applied, the wet adhesion decreased rapidly as did the conductivity. Increasing the bulk-annealing temperature did not improve adhesion (Supplementary Fig. 14), and heating above the glass transition point (T_g) of the polymer substrate led only to deformation of the substrate (Supplementary Fig. 15). When special energy sources were supplied by laser, PEDOT:PSS and the substrates underwent unconventional

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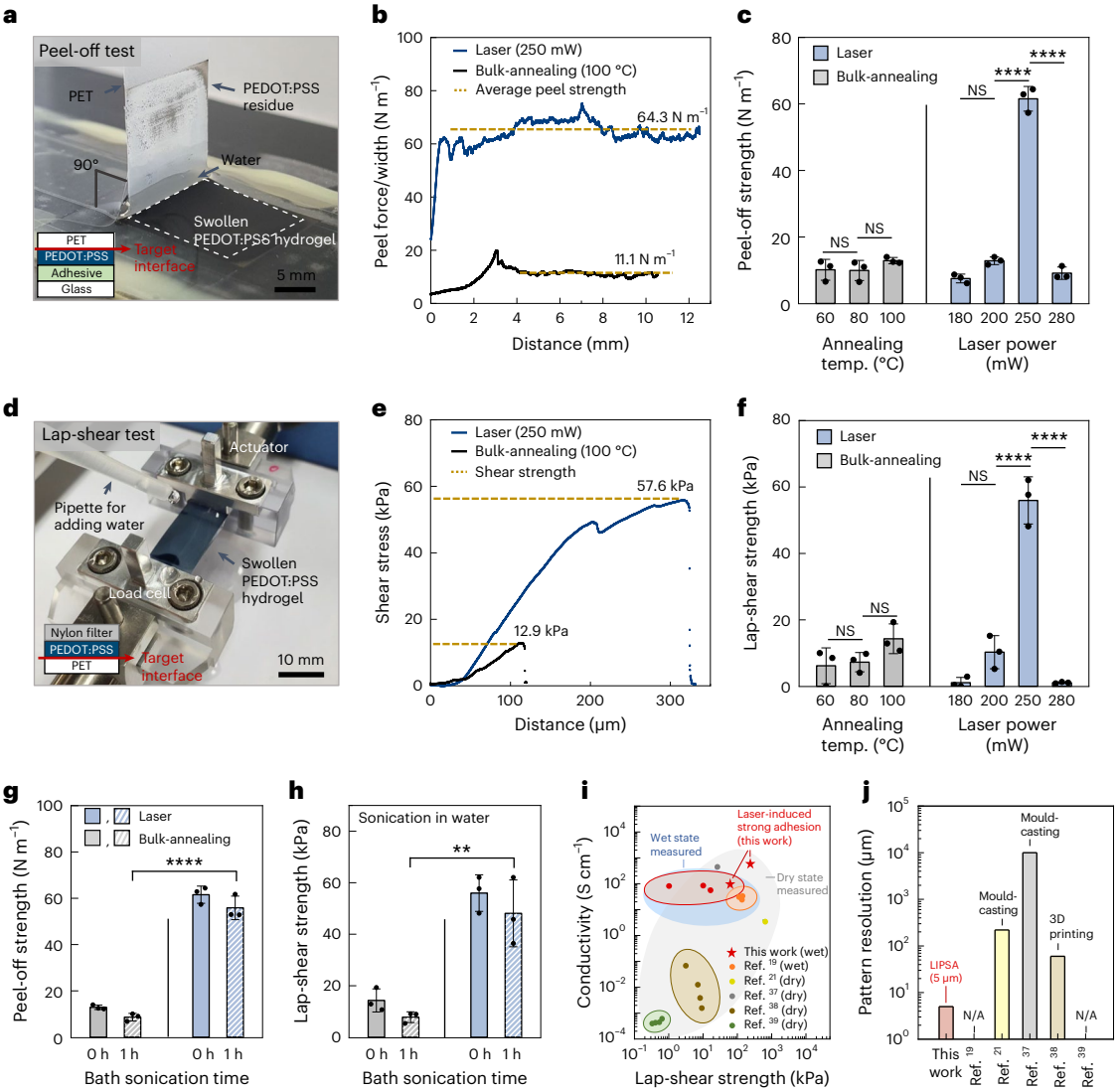


Fig. 3 | Laser-induced strong wet adhesion of conducting polymer. a, Image of 90° peel-off test set-up. **b**, Peel force–distance curve of LIPSA (250 mW) and bulk-annealing (100 °C) samples. **c**, Peel-off strength according to each fabrication condition. **d**, Image of the lap-shear test. **e**, Stress–distance curve of LIPSA (250 mW) and bulk-annealing (100 °C) samples. **f**, Lap-shear strength for each fabrication condition. **g**, Peel-off strength degradation after 1 h of bath sonication. **h**, Lap-shear strength degradation after 1 h of bath sonication. **i**, Comparison of conductivity and adhesion in the literature and in this work. The data with a grey background represent adhesion in the dry state, whereas data with a blue background represent adhesion in the wet state. **j**, Comparison of spatial resolution in previous research. Values in **c**, **f**, **g** and **h** represent means ± standard deviation ($n = 3$). The P values in **c** and **f** were calculated using one-way ANOVA followed by Tukey's multiple comparisons. In **f**, ****$P = 6.465 \times 10^{-6}$ (200–250 mW), ****$P = 1.473 \times 10^{-6}$ (250–280 mW), **$P < 0.05$ and ****$P < 0.00001$. The P values in **g** and **h** were calculated using a two-sample t test. In **g**, ****$P = 1.029 \times 10^{-5}$, and in **h**, **$P = 0.00138$, **$P < 0.05$ and ****$P < 0.00001$. 3D, three-dimensional; NS, not significant; temp., temperature.

thermodynamic events, such as rapid melting and cooling, thus minimizing substrate deformation.

Robust wet adhesion was achieved through a combination of phase separation of PEDOT:PSS and nano- and microscale changes in the roughness of the laser-treated interfaces. The water-inhibiting region contributed to the improvement of the wet adhesion by reducing water penetration. PEDOT:PSS hydrogel (LIPSA at 250 mW) in the dry state had an adhesion strength of over 227 kPa (Supplementary Fig. 16). The laser-induced local heating and cooling caused thermal expansion and contraction of the PEDOT:PSS hydrogel, resulting in microscale interface undulations. The wavy interface also contributed to the improvement of the adhesion by increasing the contact area between PEDOT:PSS and the substrate (Supplementary Note 4 and Supplementary Figs. 17–20).

Spot welding, during which PEDOT:PSS and PET melt to form a single mass, is the key to forming strong wet adhesion in LIPSA. A cross-sectional SEM image shows a smoothly welded area between them at the wavy interface (Extended Data Fig. 4a). Due to the rapid heating and cooling of the laser-irradiated area, a wavy interface is formed and local melting occurs, and PEDOT:PSS penetrates into PET in a very localized area, forming strong interlocking. A strong force is required to delaminate the PEDOT:PSS hydrogel, which results in cohesive failure of the PEDOT:PSS welded to the PET. Sulfur ions were detected in various places in SEM-energy dispersive spectroscopy images of a delaminated surface and a cross-section of laser-treated PET (Extended Data Fig. 4b,c). Since sulfur ions do not exist in PET, their presence indicates that there is a PEDOT:PSS residue that is strongly bonded to the PET. In zone 1, the PEDOT:PSS residue remains in a large lump on the wavy surface of the PET. An area with a high density of sulfur ions can be seen on the surface of the PET under the lump. In this area, the PEDOT:PSS has penetrated and welded to the PET. In zone 2, the PEDOT:PSS has been cleanly removed without leaving a residue.

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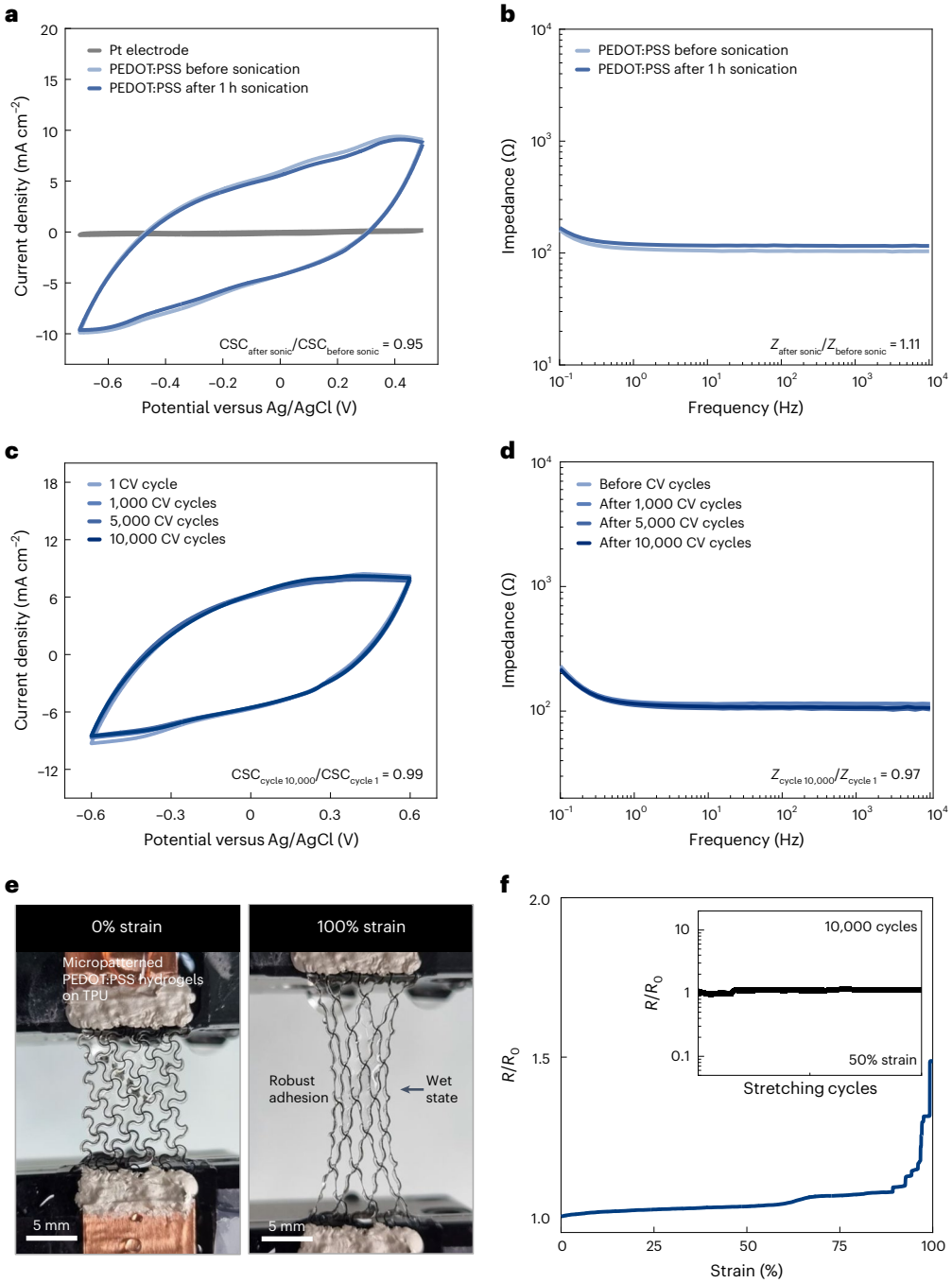


Fig. 4 | Wet adhesion and stability of PEDOT:PSS hydrogels. a, Mechanical robustness CV curve for PET treated by LIPSA before and after a 1 h sonication bath. **b**, Electrochemical impedance spectroscopy curves for PET treated by LIPSA before and after a 1 h sonication bath. **c**, Long-term CV curves for for PET treated by LIPSA. **d**, Long-term electrochemical impedance spectroscopy curves for PET treated by LIPSA. **e**, Micropatterned PEDOT:PSS hydrogels on TPU. Before stretching (left) and after stretching (right). **f**, Relative resistance change of patterned PEDOT:PSS hydrogel as a function of strain. Inset, There was no change in the resistance of the structure after 10,000 cycles of 50% strain.

However, sulfur ions were detected on the PET surface, indicating that the PEDOT:PSS had penetrated into PET. For the bulk-annealed sample, the analysis was performed in the same manner, and no penetration of PEDOT:PSS was found either on the top surface or in a cross-section (Extended Data Fig. 4c).

The effect of the laser was strongest at the interface between PEDOT:PSS and PET, causing spot welding. However, in areas not notably affected by the laser, the bond in the thickness direction between PEDOT:PSS layers was not strong, and a weak layer with low cohesive energy formed. The delamination occurred through the destruction of the weak layer within PEDOT:PSS, not at the strong interface. Therefore, EG treatment strengthened the connection of the PEDOT-rich domains and removed the weak layer, thus completing the robust wet adhesion of LIPSA (Extended Data Fig. 5).

The wet adhesion strength of PEDOT:PSS hydrogel was also evaluated through a sonication test, which is commonly used to assess the qualitative wet adhesion of polymers[18,19,36]. We also accurately measured the degradation of adhesion forces after subjecting samples to harsh ultrasonication tests. The peel and lap-shear strengths of PEDOT:PSS hydrogels (LIPSA at 250 mW and bulk-annealing at 100 °C)

were measured after the 1 h sonication test. The bulk-annealed sample exhibited a notably weakened bonding force of 8.0 N m^{-1} and 6.9 kPa, whereas the laser-treated sample showed no notable change in bonding strength, measuring 55.8 N m^{-1} and 48.1 kPa (Fig. 3g,h). These high adhesion strengths can be realized for other famous polymer substrates, including PI, TPU, SBS and SEBS, which can withstand a 1 h cavitation shock in a sonication bath with little degradation (Supplementary Fig. 6).

The fabrication of a PEDOT:PSS hydrogel with high electrical conductivity and strong bonding to the substrate in a wet state presents notable challenges. Quantitatively evaluating the wet stability of such a material is also problematic. Previous research has primarily focused on measuring the shear strength in a dry state, with only one study reporting the shear strength in a wet state (Fig. 3l). The use of crosslinked polymers to improve the strength of the bonding with the substrate results in reduced electrical performance as these polymers are insulators[21,37–39]. In contrast, pure PEDOT:PSS with a surface-treated adhesion layer demonstrates superior wet adhesion and excellent wet-state electric conductivity[19]. LIPSA, which results in outstanding wet conductivity (101 S cm^{-1}) and excellent wet adhesion (57 kPa) as well as a high pattern resolution (5 μm) (Fig. 3i,j and Supplementary Table 3), is a promising solution to these challenges.

Wet stability characterization

Due to the excellent hygroscopicity of PEDOT:PSS hydrogels, water permeates between the substrate and the PEDOT:PSS hydrogel, resulting in interfacial delamination, crack formation and detachment. These defects cause a rapid deterioration in the electrochemical performance of PEDOT:PSS hydrogels, resulting in a loss of functionality[17,40,41]. Our hydrogels exhibited exceptional electrical stability during long-term operation and under harsh environmental conditions.

The stability of PEDOT:PSS hydrogel (LIPSA at 250 mW) in wet physiological environments was evaluated in terms of electrochemical performance. The changes in CSC and impedance were measured before and after the 1 h sonication test and 10,000 cycles of the cyclic voltammetry (CV) test. Sonication tests can estimate the robust bonding of electrodes, and high-repetition CV tests can evaluate the long-term stability of electrodes. After being subjected to a 1 h sonication shock, the CSC and impedance slightly decreased: $CSC_{before\ sonic}$ = 49.85 mC cm^{-2}, $CSC_{after\ sonic}/CSC_{before\ sonic}$ = 0.95, impedance at 1 kHz was 103.9 Ω and $Z_{after\ sonic}/Z_{before\ sonic}$ = 1.11 (Fig. 4a,b). The CSC and impedance were also maintained after 10,000 cycles of the CV test: $CSC_{cycle\ 1}$ = 53.05 mC cm^{-2}, $CSC_{cycle\ 10,000}/CSC_{cycle\ 1}$ = 0.99, $Z_{cycle\ 1}$ at 1 kHz was 110.1 Ω and $Z_{cycle\ 10,000}/Z_{cycle\ 1}$ = 0.97 (Fig. 4c,d). Then, we tested the high wet stability and reusability under harsh physiological environments, imitating in vivo implant conditions. Fortunately, the differences in bonding strength between samples that were reused three and five times and a new sample that had never been reused were negligible (Supplementary Fig. 21). Both the impedance and CV were maintained well (CSC_0 = 50.04 mC cm^{-2}, CSC_5/CSC_0 = 1.01, Z_0 at 1 kHz was 210 Ω and Z_5/Z_0 = 0.89), and there was no microscopic delamination on the surface (Supplementary Fig. 22).

We characterized the stretchability and bending stability of PEDOT:PSS hydrogel electrodes (Supplementary Note 5 and Supplementary Figs. 23 and 24). To greatly enhance the stretchability of our hydrogel electrodes, we cut a TPU substrate into an identical structure as PEDOT:PSS hydrogel patterned into a serpentine shape (Fig. 4e). The relative resistance of the micropatterned PEDOT:PSS hydrogel increased by 10% at 90% strain, and it strongly adhered to the substrate even in the wet state at 100% strain. The relative resistance was maintained after 10,000 cycles of 50% strain in a cyclic stretching test (Fig. 4f). The results show that the electrochemical and electrical properties remained robust even in harsh environments, making this material suitable for long-term bioelectronic operation.

Bioelectronic applications

To verify that LIPSA is a useful fabrication process for micropatterned bioelectronics, here we describe highly robust hydrogel microelectrode arrays that can record electrophysiological signals in vivo for a long duration (Supplementary Fig. 25). Fourier-transform infrared spectroscopy confirmed that a potentially harmful EG residue was not present in the PEDOT:PSS hydrogel after washing (Supplementary Fig. 26). Then, we fabricated high-resolution implantable neural probes with 60 μm of line width and 120 μm pitches (Fig. 5a). Sixteen PEDOT:PSS hydrogel microelectrodes were patterned on a PET substrate with a relatively stiff Young's modulus for easy insertion into the rat brain. A thin polydimethylsiloxane (PDMS) layer was spin-coated for encapsulation (Fig. 5a, inset).

The fabricated flexible neural probe was then directly connected to a commercial signal acquisition system (RHS Headstages, Intan Technologies) through micro connectors (Fig. 5b and Supplementary Fig. 27). Soft neural probes with hydrogel microelectrode arrays were gently inserted into the somatosensory cortex region of the rat's brain (Fig. 5b, inset). The extracellular action potential, which a high-resolution microelectrode array can obtain, was easily recorded by our hydrogel neural probes with 30 kHz sampling rates[42]. We used high-pass filtering that ranges from 300 Hz to 7.5 kHz to precisely acquire action potentials (60 Hz of notch filter). The relative threshold of four times the root mean square level on each channel was used to detect spikes. First, we recorded the action potential of various states of the rat (Fig. 5c). In the anaesthetized state, no observable action potential was detected, but action potentials with clear spikes were periodically detected when the rat woke up. In addition, compared to spontaneous action potentials, potentials under mechanical stimuli (clamping the tail with tweezers) showed irregular and high amplitudes. Owing to the high robustness and biocompatibility of hydrogel-based neural probes produced by LIPSA, we also performed long-term signal acquisition of neural signals after 3 weeks of implantation (Fig. 5d). We recorded spontaneous action potentials and calculated the signal-to-noise ratio (SNR) for each data stream. The SNR was calculated by dividing the peak amplitude of spikes into two times the root mean square value of the noise level[43]. The SNRs were stable after implantation and even increased due to the improved contact of the electrodes to brain tissues induced by healing (Fig. 5e). The electrochemical impedance of an implanted neural probe was low enough to record neural signals after 3 weeks of recording, which shows high durability (Fig. 5f).

Since LIPSA can easily modify the geometry of PEDOT:PSS hydrogels, we also designed strain-insensitive, highly stretchable, hydrogel microelectrode arrays to record beating cardiac signals dynamically (Fig. 5g). All the fabrication procedures were the same as for the neural probes, but the substrate was replaced with TPU and an ultraviolet pulse laser was used to cut the device to the designed geometry. Six electrodes were placed on the rat's heart to record epicardial signals. The filtering frequency was modified to 1–200 Hz (60 Hz of notch filter) to obtain surface epicardial signals. After recording epicardial signals in vivo, we removed the electrode arrays. They had blood and proteins on them that were hard to wash with running water (Fig. 5h). As the microelectrode array is highly robust, we cleaned the used microelectrode array with ultrasonication. After cleaning the electrode array for 30 s in a sonication bath, it had clean surfaces and a robust appearance (Fig. 5h, right image). We then attached the device to the same rat and compared the difference in signal quality (Fig. 5i). The epicardial signal waveform and SNR showed negligible changes between the first use and second uses (Fig. 5j). The SNR of the epicardial signal was calculated by dividing the peak amplitude by the noise amplitude (Fig. 5k). The electrochemical impedance was also tested before and after the sonication tests, which showed no clear differences (Fig. 5l). This is expected to lead to reusable electronic devices, which are normally very expensive commercially.

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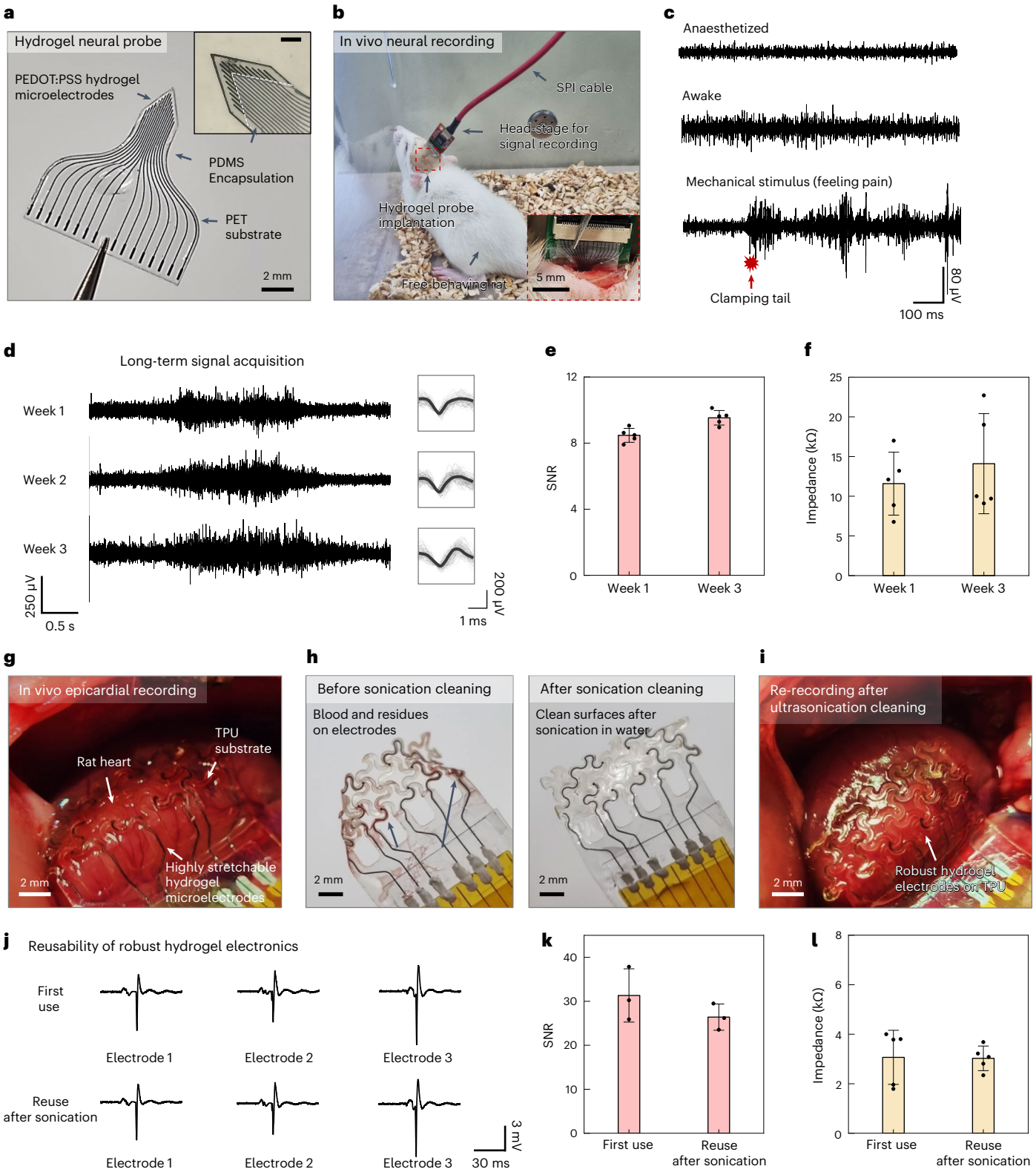


Fig. 5 | Bioelectronic applications. a, Digital image of hydrogel neural probes consisting of 16 PEDOT electrode arrays. Inset, Magnified optical image. **b**, In vivo neural recording set-up for the free-behaving rat. Inset, Implantation region for hydrogel probes. **c**, Action potential of rat's brain under various physiological states. **d**, Long-term signal acquisition using robust hydrogel neural probes. **e**, Change in SNR of action potentials after 3 weeks of implantation. **f**, Change in impedance of hydrogel neural probes after 3 weeks of implantation. **g**, Optical image of stretchable serpentine-shaped hydrogel microelectrodes on rat's heart. **h**, Robust hydrogel microelectrodes enduring ultrasonication cleaning after recording. **i**, Re-recording of epicardial signals after ultrasonic cleaning of the hydrogel microelectrodes. **j**, Reusability of robust hydrogel microelectrodes, showing little change in signal quality. **k**, Change in SNR of epicardial signals after sonication cleaning. **l**, Change in impedance of hydrogel microelectrodes after sonication cleaning. Values in **e**, **f**, **k** and **l** represent means ± standard deviation ($n = 5$ in **e**, **f** and **l** and $n = 3$ in **k**). SPI, serial peripheral interface.

Conclusions

We have described a laser-induced phase separation method to create pure PEDOT:PSS conductive hydrogels with high stability and strong wet adhesion to various polymer substrates. The PEDOT:PSS and polymer substrate are scanned by a laser, which causes partial phase separation in the PEDOT:PSS. The concentration of thermal energy at the interface creates mechanical interlocking structures with the substrate. The hydrogel is strengthened further by post-treatment with EG. The resulting pure PEDOT:PSS hydrogels show excellent mechanical stability, adhesion strength and electrical conductivity in wet conditions, including a conductivity of up to 101.4 S cm^{-1}, peel strength of 64.4 N m^{-1} and lap-shear strength of 62.1 kPa. The process can use selective laser irradiation to make high-resolution hydrogel patterns. To illustrate the potential of our method to fabricate patterned bioelectronics, we created hydrogel microelectrode arrays for in vivo electrophysiological recording, including high-resolution neural probes that collected signals from a rat brain for 3 weeks and highly stretchable and reusable cardiac probes for monitoring a rat heart. We expect our laser process could be applied to various applications requiring soft interfaces with biological systems while ensuring high electrical performance and spatial resolution.

Methods

Materials

PEDOT:PSS aqueous solution (1.0–1.3 wt% solid content, Clevios PH1000, Heraeus Electronic Materials) was passed through filters of 0.45 µm pore size, drop-casted or spin-coated on polymer substrates including PET, TPU, PI, SBS and SEBS and then dried at room temperature. To ensure stable casting of PEDOT:PSS aqueous solution on various polymer substrates, the air plasma was treated with 100 W of power for 8 min.

Pure PEDOT:PSS hydrogels fabricated by LIPSA

The laser system consists of an f-theta telecentric lens (f = 103 mm), a continuous-wave 532 nm laser (Sprout-G-5W, Lighthouse Photonics) and a Galvano mirror (hurrySCAN II, Scanlab), which was utilized to induce phase separation of PEDOT:PSS and interface structures with substrates (Supplementary Fig. 10a). To focus the laser beam tightly (Supplementary Fig. 10b), an objective lens (Mitutoyo M Plan Apo) with ×20 or ×50 magnification was used. A three-axis moving stage (Aerotech ANT130-060-XY-25DU-XY-CMS-MP-PLUS) was used for the laser scanning. Laser-scanned samples were then immersed in water to dissolve the unscanned area for patterning. After washing, samples were dried on a hot plate at 60 °C. Each dried sample was then soaked in EG for 30 min and washed with ethanol and water several times.

TEM microscopy

TEM samples were prepared using a focused ion beam. TEM images of surfaces produced by LIPSA before and after EG treatment were observed using a corrected scanning transmission electron microscope (JEOL).

Microscopy

The AFM image of the water-absorbing region was obtained by scanning the top surface where the PEDOT:PSS hydrogel was in contact with air. In contrast, the AFM image of the water-inhibiting region was obtained by scanning the bottom surface, which was in contact with the polymer substrate after the PEDOT:PSS hydrogel had delaminated from the substrate (NX10, Park Systems). SEM images of the interlocking structure were obtained by delaminating PEDOT:PSS hydrogel from the substrate. A cross-sectional image was then taken after cutting the LIPSA samples with a very sharp surgical knife (SUPRA 55VP, Carl Zeiss). Optical images were captured with an optical microscope (BX53M, Olympus).

Material characterization

The X-ray photoelectron spectroscopy analysis was performed by electron spectroscopy for chemical analysis (AXIS Supra, Kratos). The absorbance and transmittance of PEDOT:PSS and polymer substrates were measured using a ultraviolet-to-visible spectrophotometer (V-770, Jasco).

Mechanical characterization

PEDOT:PSS hydrogels were cut into a dog-bone shape with strain-tracking patterns to concentrate the stress on the gauge section using a femtosecond laser (PHAROS, L2K Co.) before being floated on the water surface. The tensile properties were measured with a water-supported laboratory-made tensile testing system comprising a load cell (LTS-50GA and 10GA, Kyowa), a high-resolution d.c. motor actuator (M-111.1 DG, PI) and a CCD camera (F-145Bs IRF, Marlin). A PDMS-coated grip held free-standing PEDOT:PSS hydrogel and a uniaxial tensile load were applied with a displacement of 1% strain per minute. The load cell read the tensile load, and a digital image correlation system measured the strain by tracking patterns. The tensile test was performed without a drag force while the specimen contained sufficient water (Supplementary Fig. 28).

Adhesion characterization

PEDOT:PSS hydrogels with dimensions of 1 cm in width and 1.5 cm in length were fabricated on a 50 µm PET substrate by LIPSA and bulk-annealing. The peel-off strength was measured with a laboratory-made 90° peel-off test system consisting of a load cell (SMA-60N, Interface) and two linear stages (M-413.3PD, PI, and LS-65 2″ DC MLS, PI) that were synchronized to maintain a peeling angle of 90°. After applying epoxy adhesive (DP420, 3M) between the glass substrate and PEDOT:PSS, the sample was cured for 12 h in a convection oven at 40 °C to ensure complete bonding. The PET substrate was peeled off from PEDOT:PSS in water at an angle of 90° to measure the peel strength. The PET substrate was attached to the load cell using 3M magic tape, and the force at the moment of detachment was measured by pulling it along the z-axis at a speed of 0.03 mm s^{-1}. The lap-shear strength was measured by a high-precision micromechanical test system (Delaminator Adhesion Test System, DTS Co.). A lap-shear test of hydrate PEDOT:PSS and polymer substrate was performed using a nylon membrane filter (pore size 0.45 µm; SciLab) as a water-permeable back plate. A nylon membrane filter was placed on the PEDOT:PSS, soaked in water for 24 h and pressed at 1 MPa and 50 °C for 10 min using a hot press. The hot-pressing strongly bonded the nylon membrane filter to the laser-treated PEDOT:PSS. After bonding, the sample had a shear strength of 227 kPa in a sufficiently dry state, and there was a cohesive fracture of the nylon filter. After mounting the bonded specimen onto a high-precision micromechanical test system, water was supplied to a nylon filter to keep the PEDOT:PSS sufficiently wet. The shear strength of hydrated PEDOT:PSS hydrogel was measured by pulling the PET and the nylon membrane filter at a speed of 1 µm s^{-1}.

Swelling behaviour characterization

The thickness of the PEDOT:PSS hydrogel was measured using an optical microscope (LV-UEPI-N, Nikon). After fabricating the PEDOT:PSS hydrogel on the PET substrate, it was cut vertically from the PET side using a surgical knife. The PEDOT:PSS hydrogel was immersed in water for 24 h to absorb sufficient water, and the PET substrate was fixed with an aluminium block to observe a cross-section. The thickness of the PEDOT:PSS hydrogel was regarded as the distance from the PET/PEDOT:PSS hydrogel interface to the top side of the cross-sectional image. The measured swollen PEDOT:PSS hydrogel specimens were dried to measure the dry thickness. After drying the specimens, the thickness was measured with the same procedure. The swelling ratio was calculated by dividing the thickness in the hygroscopic expansion state by the thickness in the dry state.

Electrical characterization
The sheet resistance (R_s) of each sample was assessed using a four-point probe (2400, Keithley). The electrical conductivity was then calculated: $\sigma = 1/R_s t$, where σ is the electrical conductivity and t is the thickness of the PEDOT:PSS hydrogel. A laboratory-made stretching system consisting of Keithley and linear stages was used to measure the change in the resistance of the conductive hydrogels.

Electrochemical characterization
The electrochemical properties (impedance and CSC) were measured using a potentiostat (VersaSTAT 3, Princeton Applied Research). Conductive hydrogels were immersed in phosphate-buffered saline for 1 h before being measured.

Wet adhesion stability characterization
To evaluate the wet stability in a physiological environment, pure PEDOT:PSS hydrogels were subjected to ultrasonication in vials containing phosphate-buffered saline for 1 h. Adhesion measurements and electrochemical characterization were performed after sonication.

Imitating an in vivo test
In a simplified experiment, the hydrogels were repeatedly soaked with animal blood and washed, and then the adhesive strength was measured. Blood agar was mixed with defibrinated sheep blood to simulate an in vivo implant environment with many foreign substances and proteins. The conductive hydrogel on PET absorbed this blood and was then dried. Afterwards, it was immersed in water and subjected to ultrasonic cleaning for 10 min. The process of absorbing and rinsing was repeated for five cycles.

Animal husbandry
Five-week-old male Sprague Dawley rats were prepared from OrientBio (Seongnam, South Korea). In vivo experiments were conducted with the approval of the Konkuk University Institutional Animal Care and Use Committee (KU23045). All animals were maintained in a 12 h light and 12 h dark cycle at 23 ± 1 °C and 50 ± 10% relative humidity with free access to food and water.

Neural probe implantation
The Sprague Dawley rats were anaesthetized with intraperitoneal administration of ketamine (50 mg kg^{-1}, Yuhan) and xylazine (10 mg kg^{-1}, Rompun, Bayer Korea). An incision was made in the skin at the head region and four burr holes were made using a mini drill (DS-1800, Dongsung Electronics) with 0.5 mm burr to anchor the screw into the skull. A 2 mm slit was made in the skull above the somatosensory cortex region for the implantation of the electrode. The electrode was inserted and secured to the skull surface with dental cement (Vertex Self-Curing Powder, Vertex Dental). Following implantation, aspirin (100 mg kg^{-1}) was administered every 24 h for 72 h postoperatively for analgesia.

Implantation of the epicardial recording device
Sprague Dawley rats were anaesthetized with isoflurane. Electrophysiological heart signals were then recorded from the heart surface. Anaesthesia was induced in 5% isoflurane mixed with 100% oxygen, followed by endotracheal intubation with a catheter. Anaesthesia was maintained in 2.5% isoflurane mixed with 100% oxygen, and ventilation was supported by a respirator throughout the procedure. The skin and rib were incised to expose the heart, and a sensor was attached to the surface of the heart.

Reporting summary
Further information on the research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

D.W. and HJ.K. designed and carried out the projects. H.K. and J.C. designed and performed the MD simulations. J.K. and M.W.K. carried out the in vivo experiments and were advised by C.-Y.K. J.A. performed the heat transfer simulations. K.M. and S.H. set the optical system for high resolution and performed the experiments. Y.L. performed the imaging characterizations. T.-S.K. and S.H.K. supervised the research programme. All authors discussed the results and jointly wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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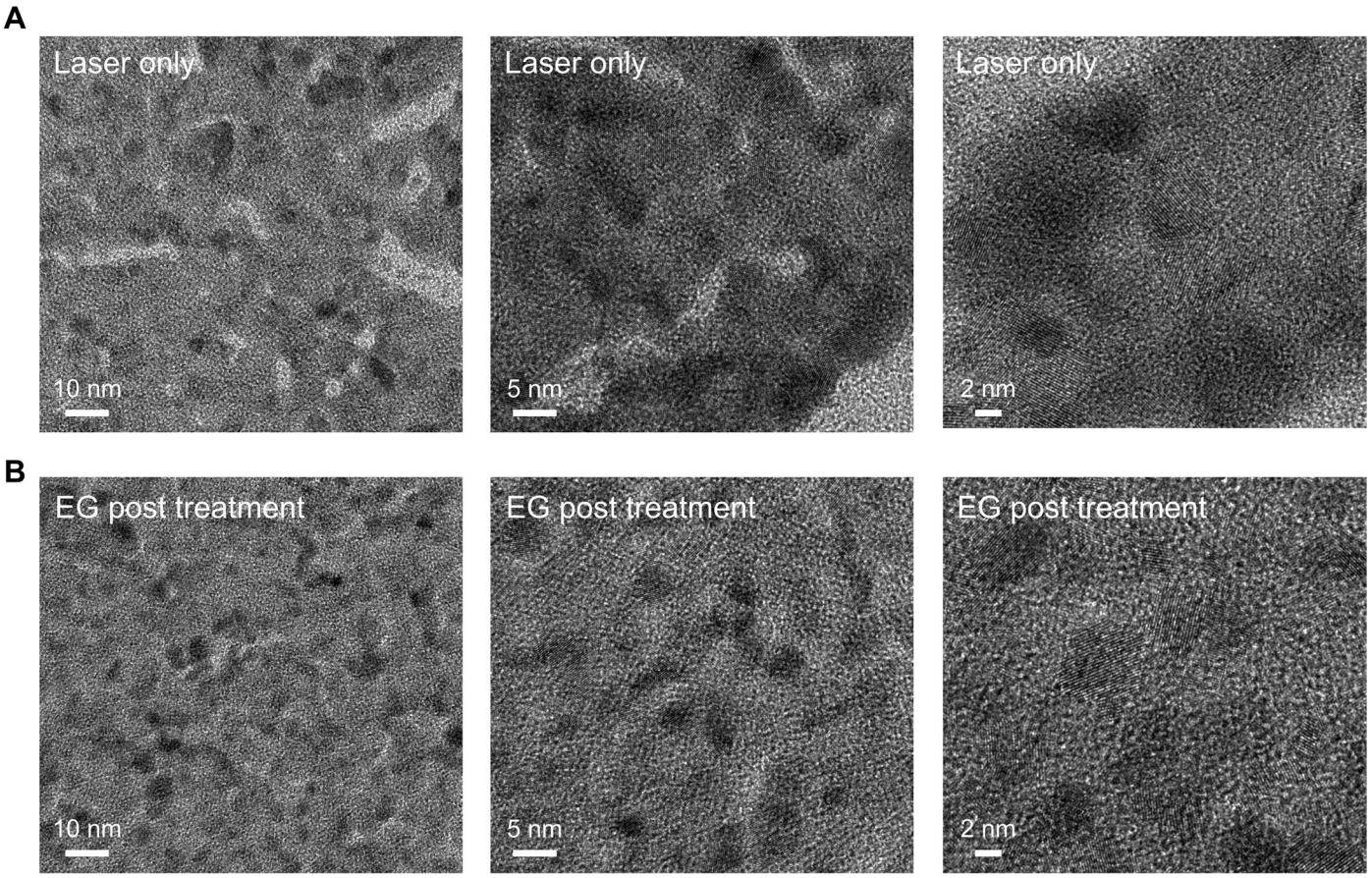
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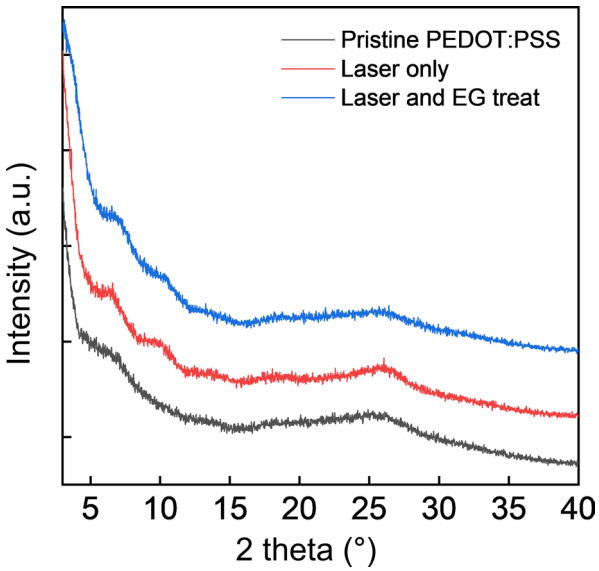
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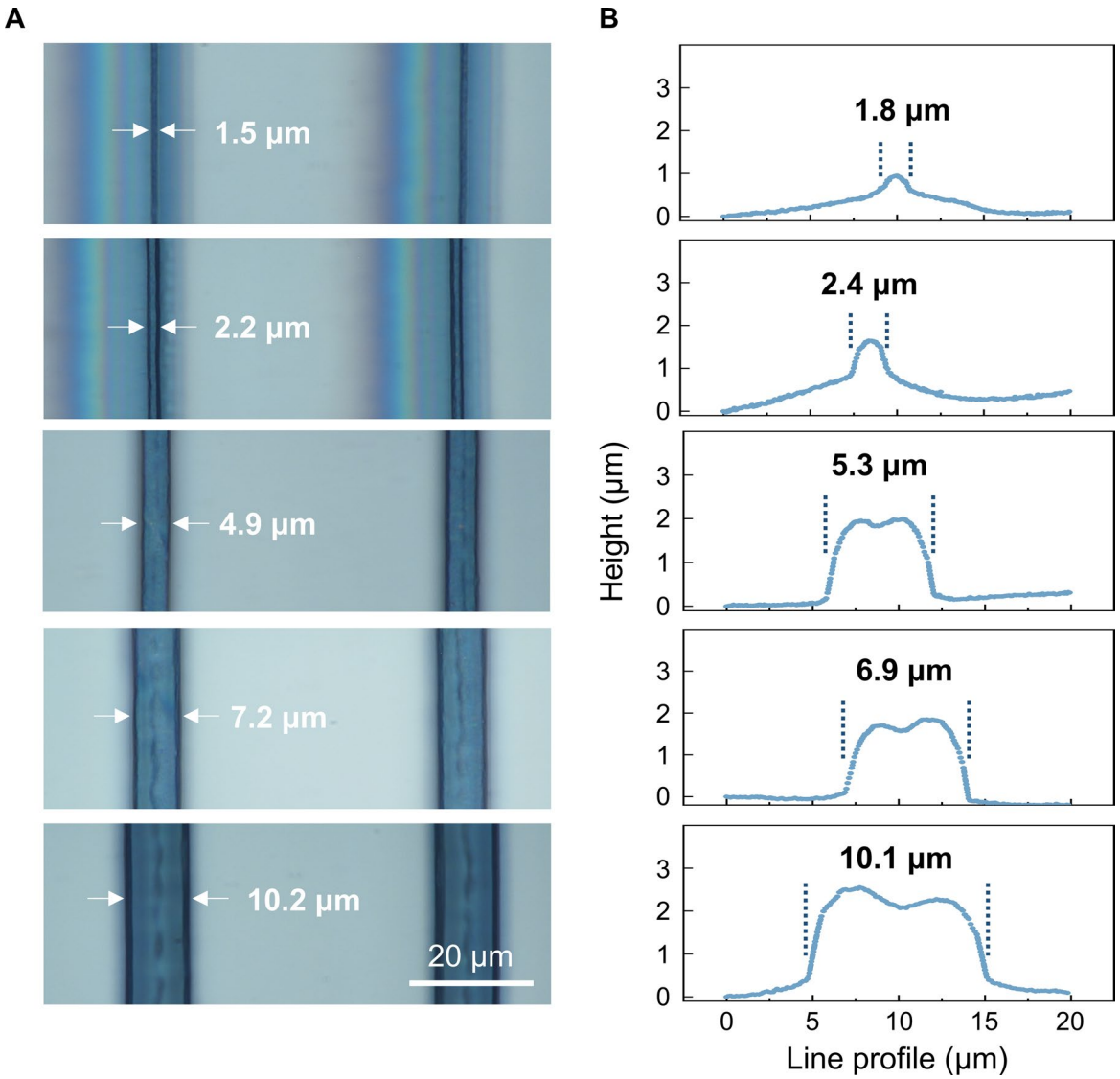
Extended Data Fig. 1 | TEM image of PEDOT:PSS surface by LIPSA before and after EG treatment. (**A**) TEM images of PEDOT:PSS before EG treatment. (**B**) TEM images of PEDOT:PSS after EG treatment.

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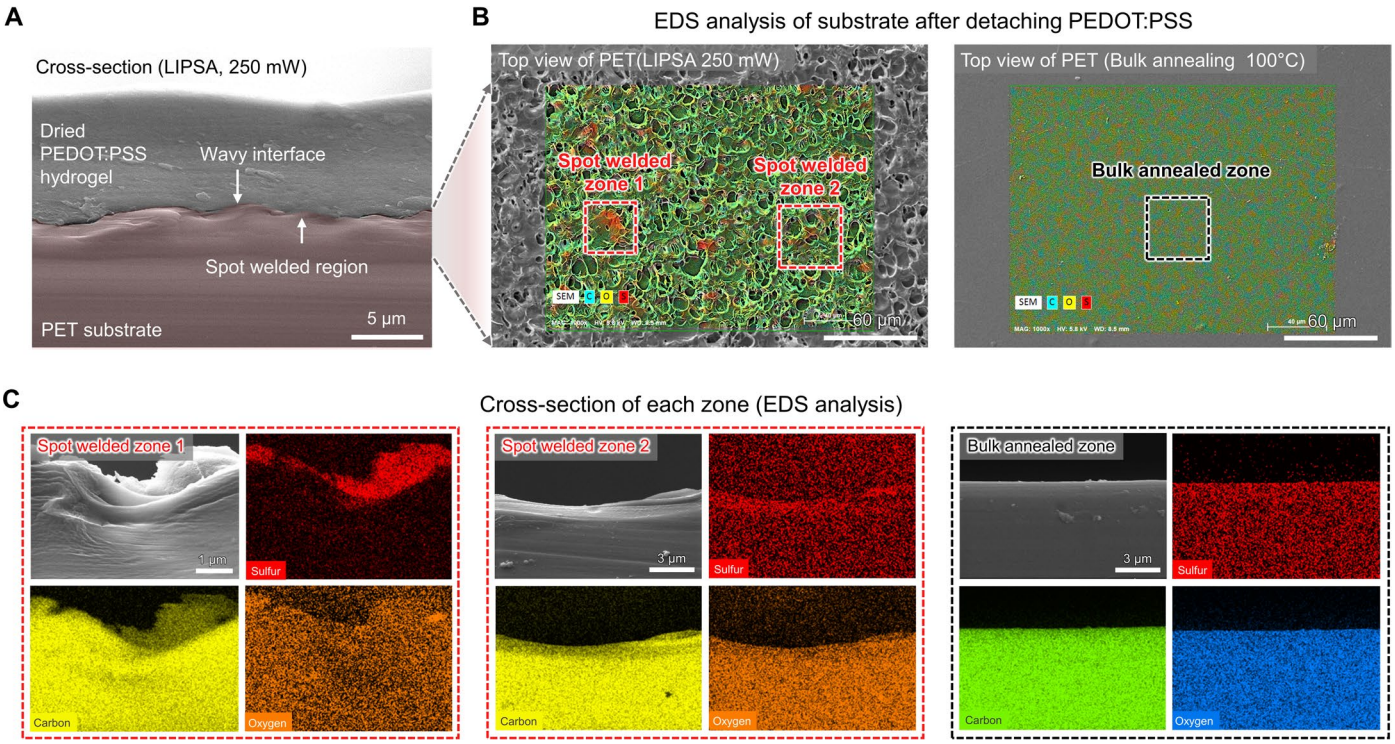
Extended Data Fig. 2 | XRD data of pristine PEDOT:PSS, laser-treated PEDOT:PSS before and after EG treatment. No distinct difference in XRD peaks is found after LIPSP.

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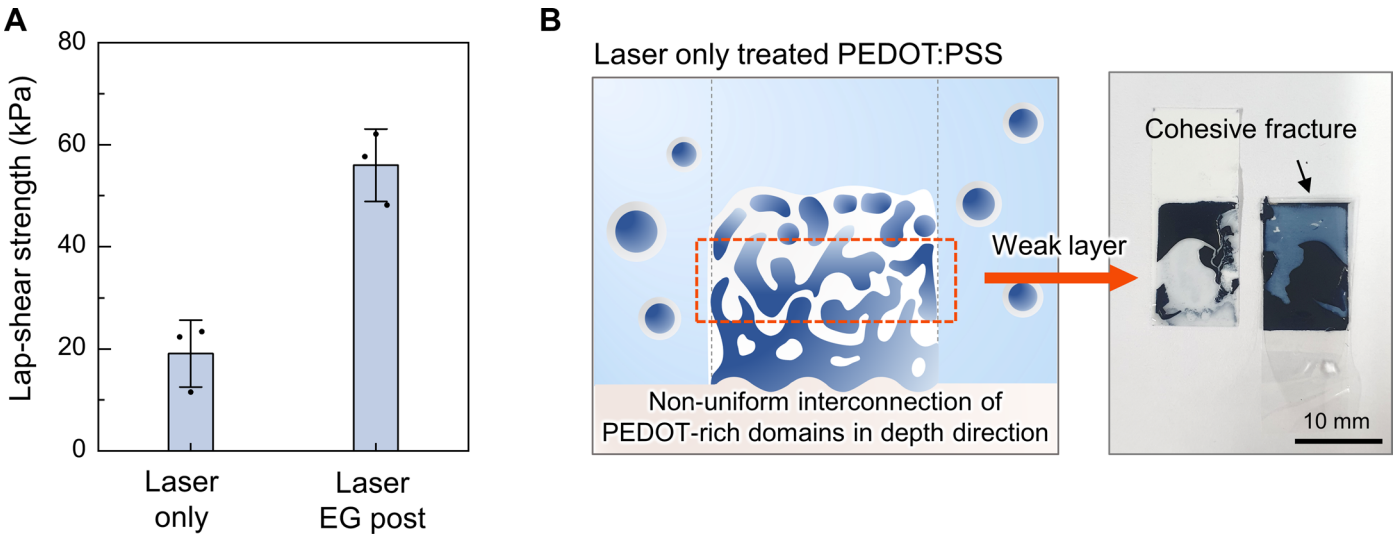
Extended Data Fig. 3 | High-resolution patterning results using a tightly focused laser beam by the objective lens. (**A**) Optical microscope images of dried PEDOT:PSS hydrogel patterns, (**B**) AFM topography of dried PEDOT:PSS hydrogels patterns.

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Extended Data Fig. 4 | Characterization of the PET substrates after laser process and bulk annealing. (**A**) Cross-sectional SEM image of PEDOT:PSS and PET interfaces. (**B**) EDS analysis of interfaces on PET-side after delaminating PEDOT:PSS. (**C**) EDS analysis on the interfaces on the PET side in cross-sectional view.

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Extended Data Fig. 5 | Effect of EG post-treatment on adhesion force of PEDOT:PSS hydrogels to substrates. (**A**) Lap-shear strength of PEDOT:PSS hydrogels before and after EG treatments. (**B**) The weak layer generated inside PEDOT:PSS causes cohesive fractures.

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